

Unix Origin and Basic Commands

What is unix?

Unix is a command user interface. It can be able to work on any type of computers. (microcomputers, mainframe or on any device that's why the name suggests Command User Interface.)

- This was developed by Ken Thompson and Dennis Ritchie in 1969.
- This was the first OS developed and based on this OS we are now able to enjoy so many OS.
- Unix is based on C language and was developed by Dennis Ritchie.

Salient features of unix

- Multiuser (same computer resources – hard disk etc. are available to many users.)
- Multitasking (open multiple shells at one time and do multiple work simultaneously)
- Communication ftp, ssh etc.
- Secured
- Portability (Highly Portable)

Flavours of Unix

As UNIX is open source, multiple flavours are available with lot of extensions and improvements.

- Ubuntu
- RedHat
- Centos
- Fedora
- Slackware
- open solaris

- Suse Linux Enterprise server (SLES)
- Open Suse

Unix system organization

*The **Shell** or command interpreter interprets the command which we give on screen and then sends signal to the **kernel** which ultimately executes them.*

Shell:

- * It is the outer layer of UNIX operating System.
- * It reads our command, verify syntax and check whether the corresponding command related application is available or not.
- * If everything is proper, then shell interprets our command into kernal understandable form and handover to the kernal.
- * Shell acts as interface between user and kernal.

Kernal:

- * It is the core component of UNIX operating system.
- * It is responsible to execute our commands.
- * It is responsible to interact with hardware components.
- * Memory allocation and processor allocation will taken care by kernal.

How it Works

1) User types the command in the terminal.

touch durga.txt

2) Shell reads that command. It will check whether that command is valid or not and whether it is used properly or not. If everything is proper, then shell

interprets/translates that command into kernel understandable form.

3) Shell handovers that interpreted command to the kernel.

4) Kernel executes that command and perform required activity.

5) Once command execution completed, then shell returns unix prompt (\$ OR # OR %).

6) \$ OR # OR % represents it is ready for the next command

Types of Shells

- Bourne Shell(Bash and currently we will be working on this only)
- C shell
- Korn Shell

The Most commonly used Basic Commands:

- 1) pwd - Print working directory
- 2) ls - List our all files and directories
- 3) mkdir - Create directory
- 4) cd - Change directory
- 5) touch - To create a file
- 6) rmdir - Remove directory
- 7) rm - To remove file
- 8) cal - Display Monthly Calander
- 9) date - Display current date and time.
- 10) help - To display list of commands.
- 11) hello - To display brief system information.
- 12) clear - To clear terminal.
- 13) exit - To logout session.